1 Question 1 — Brief Answers [15 pts]

Part (a) — TRUE / FALSE [3 pts] (1 mark each)

	Statement	Answer	Reason	Variants
i	Backpropagation *properly implemented* will *always* yield global minima.	**FALSE**	BP performs gradient descent on a generally non-convex loss; converges to a local minimum or saddle point, not necessarily global.	V1, V3
i	Backpropagation is effective *only* for convex optimization.	**FALSE**	BP is widely used and effective for non-convex neural network training; convexity is not required.	V2
i	Backpropagation will *never* yield global minima.	**FALSE**	For convex losses (e.g. a single-layer linear network with MSE), BP *can* find a global minimum. The claim "never" is too strong.	V4
ii	BP works well when weights are initialized as *zeros*.	**FALSE**	Zero initialization causes symmetry: all hidden units receive identical gradients and learn identical features (symmetry-breaking problem).	V1, V3
ii	XOR cannot be implemented with a single-layer network.	**TRUE**	XOR is not linearly separable; a single-layer perceptron/SLP cannot implement it.	V2, V4
iii	Residual/skip connections help in reducing the vanishing gradient problem in deep NNs.	**TRUE**	Skip connections provide gradient highways (identity shortcut), alleviating vanishing gradients.	V1
iii	Adding an L_1 norm of weights to the loss leads to sparsity.	**TRUE**	L_1 regularization introduces a sub-gradient of ± 1 at non-zero weights, promoting exact zeros in the solution (Lasso effect).	V2
iii	Cross entropy is popular for *regression* problems.	**FALSE**	Cross entropy is used for *classification*. MSE / MAE are standard regression losses.	V3
iii	Dropout is a method for reducing the number of parameters and getting smaller networks (model compression).	**FALSE**	Dropout is a *regularization* technique (randomly zeroes activations during training); it does not reduce parameters at inference or compress the model.	V4

Part (b) — Gradient Descent Update Equations [3 pts] (1 mark each, common to all variants)

(b-i) Simple Gradient Descent [1 pt]

$$\mathbf{w}^{k+1} = \mathbf{w}^{k} - \eta\,\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}^{k})$$

Symbols: $\eta > 0$ is the learning rate (step size); $\nabla_{\mathbf{w}}\mathcal{L}$ is the gradient of the loss w.r.t. $\mathbf{w}$.

Rubric:

- 1 pt: correct update direction ($-\eta\nabla$), correct superscript k and $k+1$, symbols defined.
- 0.5 pt: correct direction but symbols not defined or minor notational issue.
- 0 pt: wrong sign or direction.

(b-ii) Gradient Descent with Momentum [1 pt]

Convention 1 (velocity update):

$$\mathbf{v}^{k+1} = \beta\,\mathbf{v}^{k} + \nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}^{k}), \qquad \mathbf{w}^{k+1} = \mathbf{w}^{k} - \eta\,\mathbf{v}^{k+1}$$

Convention 2 (update with damped step):

$$\mathbf{v}^{k+1} = \beta\,\mathbf{v}^{k} - \eta\,\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}^{k}), \qquad \mathbf{w}^{k+1} = \mathbf{w}^{k} + \mathbf{v}^{k+1}$$

Symbols: $\beta \in [0, 1)$ is the momentum coefficient; $\mathbf{v}^{k}$ is the velocity/momentum term (initialized to $\mathbf{0}$).

Either convention is acceptable. Award full marks for either, provided symbols are defined.

Rubric:

- 1 pt: both update equations present, symbols defined, either convention.
- 0.5 pt: one equation correct / symbols missing.
- 0 pt: wrong or only plain GD given.

(b-iii) Newton's Method [1 pt]

$$\mathbf{w}^{k+1} = \mathbf{w}^{k} - \left[\mathbf{H}^{k}\right]^{-1}\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}^{k})$$

Symbols: $\mathbf{H}^{k} = \nabla^2_{\mathbf{w}}\mathcal{L}(\mathbf{w}^{k})$ is the Hessian matrix of the loss evaluated at $\mathbf{w}^{k}$.

Rubric:

- 1 pt: Hessian inverse present, correct update, defined.
- 0.5 pt: correct form without inverse or Hessian not named.
- 0 pt: wrong expression.

Part (c) — Z-score Normalization [3 pts] (one per variant)

Method: $z_i = \dfrac{x_i - \mu}{\sigma}$ where μ is the mean and σ is the standard deviation.

Two conventions are both correct:

- **Population std:** $\sigma^2 = \dfrac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2$
- **Sample std:** $s^2 = \dfrac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu)^2$

Award full marks for either, provided the computation is consistent.

Variant 1: $D = \{7, 8, 10, 11, 11, 13\}$

$\mu = \dfrac{7+8+10+11+11+13}{6} = \dfrac{60}{6} = 10$

Deviations: $-3,\ -2,\ 0,\ +1,\ +1,\ +3$. Sum of squares $= 9+4+0+1+1+9 = 24$.

Population std: $\sigma = \sqrt{24/6} = \sqrt{4} = 2$

$$D_{\text{norm}} = \left\{-\tfrac{3}{2},\ -1,\ 0,\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{3}{2}\right\} = \{-1.5,\ -1,\ 0,\ 0.5,\ 0.5,\ 1.5\}$$

Sample std: $s = \sqrt{24/5} = \sqrt{4.8} \approx 2.191$

$$D_{\text{norm}} \approx \{-1.369,\ -0.913,\ 0,\ 0.456,\ 0.456,\ 1.369\}$$

Variant 2: $D = \{14, 16, 19, 23, 23, 25\}$

$\mu = \dfrac{14+16+19+23+23+25}{6} = \dfrac{120}{6} = 20$

Deviations: $-6,\ -4,\ -1,\ +3,\ +3,\ +5$. Sum of squares $= 36+16+1+9+9+25 = 96$.

Population std: $\sigma = \sqrt{96/6} = \sqrt{16} = 4$

$$D_{\text{norm}} = \left\{-\tfrac{6}{4},\ -1,\ -\tfrac{1}{4},\ \tfrac{3}{4},\ \tfrac{3}{4},\ \tfrac{5}{4}\right\} = \{-1.5,\ -1,\ -0.25,\ 0.75,\ 0.75,\ 1.25\}$$

Sample std: $s = \sqrt{96/5} = \sqrt{19.2} \approx 4.382$

$$D_{\text{norm}} \approx \{-1.370,\ -0.913,\ -0.228,\ 0.685,\ 0.685,\ 1.142\}$$

Variant 3: $D = \{43, 45, 49, 51, 55, 57\}$

$\mu = \dfrac{43+45+49+51+55+57}{6} = \dfrac{300}{6} = 50$

Deviations: $-7,\ -5,\ -1,\ +1,\ +5,\ +7$. Sum of squares $= 49+25+1+1+25+49 = 150$.

Population std: $\sigma = \sqrt{150/6} = \sqrt{25} = 5$

$$D_{\text{norm}} = \{-1.4,\ -1,\ -0.2,\ 0.2,\ 1,\ 1.4\}$$

Sample std: $s = \sqrt{150/5} = \sqrt{30} \approx 5.477$

$$D_{\text{norm}} \approx \{-1.278,\ -0.913,\ -0.183,\ 0.183,\ 0.913,\ 1.278\}$$

Variant 4: $D = \{93, 94, 100, 102, 105, 106\}$

$\mu = \dfrac{93+94+100+102+105+106}{6} = \dfrac{600}{6} = 100$

Deviations: $-7,\ -6,\ 0,\ +2,\ +5,\ +6$. Sum of squares $= 49+36+0+4+25+36 = 150$.

Population std: $\sigma = \sqrt{150/6} = \sqrt{25} = 5$

$$D_{\text{norm}} = \{-1.4,\ -1.2,\ 0,\ 0.4,\ 1.0,\ 1.2\}$$

Sample std: $s = \sqrt{150/5} = \sqrt{30} \approx 5.477$

$$D_{\text{norm}} \approx \{-1.278,\ -1.095,\ 0,\ 0.365,\ 0.913,\ 1.095\}$$

Rubric for (c):

- 3 pts: correct mean, correct std (either convention), correct normalized values.
- 2 pts: correct mean and std but arithmetic error in final values.
- 1 pt: correct mean only, or correct method but wrong μ or σ.
- 0 pts: wrong method or no attempt.

Part (d) — Test Your Breadth [3 pts] (1 mark each)

Question	Accepted Answer(s)	Variant
Nobel Prize (Physics) 2024 for neural networks	**Geoffrey Hinton** (and/or John Hopfield). Hinton received the Physics Nobel for foundational contributions to ANNs.	V1
Expand **RNN**	**Recurrent Neural Network**	V1
Popular open-source LLM	**LLaMA / Llama 2 / Llama 3** (Meta), **Mistral**, **Falcon**, **Gemma**, **DeepSeek**, or any credible open-source LLM.	V1
Indian-origin Turing Award 1994 (Hyderabad connection)	**Raj Reddy** (Dabbala Rajagopal Reddy). Turing Award 1994 for AI; born in Andhra Pradesh, strong IIIT Hyderabad connection.	V2
Expand **CNN**	**Convolutional Neural Network**	V2
Use case of "Nano Banana"	Image generation (if someone says non-existent, give credits because of spelling mistake)	V2
AI researcher who left Meta and founded startup; pioneering in CNNs	**Yann LeCun** is the canonical answer for CNN contributions. Accept any well-known researcher who fits the description (news may have changed after training cutoff). Also accept **Soumith Chintala** or equivalent with justification.	V3
Expand **ReLU**	**Rectified Linear Unit**	V3
Primary GPU manufacturer for deep learning	**NVIDIA**	V3
Award given to Demis Hassabis and John Jumper in 2024	**Nobel Prize in Chemistry** (for AlphaFold protein structure prediction, 2024).	V4
Expand **GPU**	**Graphics Processing Unit**	V4
Commercial LLM strong in coding	**Claude** (Anthropic), **GPT-4o / ChatGPT**, **Gemini**, **DeepSeek-Coder**, **GitHub Copilot** (Codex), or any credible coding-focused commercial model.	V4

Rubric for (d): 1 mark per sub-item; use professional judgment; one clearly correct answer is sufficient.

Part (e) — Learnable Parameters in MLP (no bias) [3 pts]

Formula: Parameters = (inputs $\times$ hidden) + (hidden $\times$ outputs)

Variant	Inputs	Hidden	Outputs	Params	Computation
V1	3	5	1	**20**	$3 \times 5 + 5 \times 1 = 15 + 5$
V2	4	3	2	**18**	$4 \times 3 + 3 \times 2 = 12 + 6$
V3	2	8	1	**24**	$2 \times 8 + 8 \times 1 = 16 + 8$
V4	7	5	1	**40**	$7 \times 5 + 5 \times 1 = 35 + 5$

Rubric for (e):

- 3 pts: correct final answer with correct computation.
- 2 pts: correct formula applied but arithmetic error.
- 1 pt: partial formula (only one layer counted).
- 0 pts: wrong approach.

Note: If student includes biases: V1→26, V2→23, V3→33, V4→46. Deduct 1 pt but do not penalise twice if clearly stated.

2 Question 2 — MLP Forward Pass, Backpropagation, SLP Equivalence [15 pts]

Network Architecture (all variants)

From Fig. 1, the two-input two-hidden-neuron one-output MLP computes:

$$
\begin{aligned}
p_1 &= w_{11}x_1 + w_{12}x_2, & q_1 &= \phi_1(p_1) = \max(0, p_1)\\
p_2 &= w_{21}x_1 + w_{22}x_2, & q_2 &= \phi_1(p_2) = \max(0, p_2)\\
s &= u_1q_1 + u_2q_2, & o &= \phi_2(s) = s \quad \text{(linear)}\\
\mathcal{L} &= \tfrac{1}{2}(y - o)^2
\end{aligned}
$$

Chain Rule (common to all variants) [3 pts]

$$\frac{\partial \mathcal{L}}{\partial w_{22}} = \frac{\partial \mathcal{L}}{\partial o} \cdot \frac{\partial o}{\partial s} \cdot \frac{\partial s}{\partial q_2} \cdot \frac{\partial q_2}{\partial p_2} \cdot \frac{\partial p_2}{\partial w_{22}} = (o - y) \cdot 1 \cdot u_2 \cdot \mathbf{1}[p_2 > 0] \cdot x_2$$

where $\mathbf{1}[p_2 > 0]$ is the ReLU derivative (1 if $p_2 > 0$, else 0).

Rubric (b):

- 3 pts: all five partial derivative terms present in correct order.
- 2 pts: four terms correct (e.g. missing $\partial o/\partial s = 1$ is acceptable if implied).
- 1 pt: chain rule set up but key terms missing/wrong.
- 0 pts: no chain rule.

Variant 1

Given: $\mathbf{W} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $y = 2$.

(a) Forward Pass [3 pts]

$$p_1 = 1 \cdot 1 + 1 \cdot 1 = 2, \quad p_2 = 1 \cdot 1 + 1 \cdot 1 = 2$$

$$q_1 = \text{ReLU}(2) = 2, \quad q_2 = \text{ReLU}(2) = 2$$

$$s = 2 \cdot 2 + 1 \cdot 2 = 6, \quad \boxed{o = 6}$$

(c) Numerical Gradient [4 pts]

$$\frac{\partial \mathcal{L}}{\partial o} = o - y = 6 - 2 = 4$$

$$p_2 = 2 > 0 \;\Rightarrow\; \mathbf{1}[p_2 > 0] = 1$$

$$\frac{\partial \mathcal{L}}{\partial w_{22}} = (o - y) \cdot u_2 \cdot \mathbf{1}[p_2 > 0] \cdot x_2 = 4 \cdot 1 \cdot 1 \cdot 1 = \boxed{4}$$

(d) SLP Equivalence (linear ϕ_1) [5 pts]

If ϕ_1 is the identity, then:

$$o = \mathbf{u}^\top \mathbf{W}\mathbf{x} = (\mathbf{W}^\top \mathbf{u})^\top \mathbf{x}$$

$$\mathbf{v} = \mathbf{W}^\top \mathbf{u} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+1 \\ 2+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

The equivalent SLP has weights $v_1 = 3$ from x_1 and $v_2 = 3$ from x_2.

SLP diagram: Two inputs x_1, x_2 feeding into a single neuron with weights 3 and 3 respectively, producing output $y = 3x_1 + 3x_2$.

Variant 2

Given: $\mathbf{W} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $y = 4$.

(a) Forward Pass [3 pts]

$$p_1 = (-1)(1) + (1)(1) = 0, \quad p_2 = (-1)(1) + (1)(1) = 0$$

$$q_1 = \mathrm{ReLU}(0) = 0, \quad q_2 = \mathrm{ReLU}(0) = 0$$

$$s = (-1)(0) + (3)(0) = 0, \qquad \boxed{o = 0}$$

(c) Numerical Gradient [4 pts]

$$\frac{\partial \mathcal{L}}{\partial o} = o - y = 0 - 4 = -4$$

$p_2 = 0$: ReLU derivative is **undefined**; accept either convention:

- **Convention A** ($\phi_1'(0) = 0$): $\dfrac{\partial \mathcal{L}}{\partial w_{22}} = (-4) \cdot 3 \cdot 0 \cdot 1 = \boxed{0}$
- **Convention B** ($\phi_1'(0) = 1$): $\dfrac{\partial \mathcal{L}}{\partial w_{22}} = (-4) \cdot 3 \cdot 1 \cdot 1 = \boxed{-12}$

Either answer (0 or -12) earns full marks provided the reasoning is consistent with a stated convention.

(d) SLP Equivalence (linear ϕ_1) [5 pts]

$$\mathbf{v} = \mathbf{W}^\top \mathbf{u} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} (-1)(-1) + (-1)(3) \\ (1)(-1) + (1)(3) \end{bmatrix} = \begin{bmatrix} 1-3 \\ -1+3 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

SLP weights: $v_1 = -2$, $v_2 = 2$. Output: $o = -2x_1 + 2x_2$.

Variant 3

Given: $\mathbf{W} = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$, $y = 3$.

(a) Forward Pass [3 pts]

$$p_1 = (-1)(-1) + (1)(-1) = 1 - 1 = 0$$
$$p_2 = (-2)(-1) + (1)(-1) = 2 - 1 = 1$$
$$q_1 = \text{ReLU}(0) = 0, \quad q_2 = \text{ReLU}(1) = 1$$
$$s = 2 \cdot 0 + 2 \cdot 1 = 2, \qquad \boxed{o = 2}$$

$p_1 = 0$: accept $q_1 = 0$ regardless of ReLU convention since ReLU(0)=0 by all conventions.

(c) Numerical Gradient [4 pts]

$$\frac{\partial \mathcal{L}}{\partial o} = o - y = 2 - 3 = -1$$

$p_2 = 1 > 0 \ \Rightarrow\ \mathbf{1}[p_2 > 0] = 1$

$$\frac{\partial \mathcal{L}}{\partial w_{22}} = (-1) \cdot 2 \cdot 1 \cdot x_2 = -2 \cdot (-1) = \boxed{2}$$

(e) SLP Equivalence (linear ϕ_1) [5 pts]

$$\mathbf{v} = \mathbf{W}^\top \mathbf{u} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} (-1)(2) + (-2)(2) \\ (1)(2) + (1)(2) \end{bmatrix} = \begin{bmatrix} -2-4 \\ 2+2 \end{bmatrix} = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$$

SLP weights: $v_1 = -6$, $v_2 = 4$. Output: $o = -6x_1 + 4x_2$.

Variant 4

Given: $\mathbf{W} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, $y = -2$.

(a) Forward Pass [3 pts]

$$p_1 = (1)(0) + (-1)(2) = -2, \quad p_2 = (1)(0) + (-1)(2) = -2$$
$$q_1 = \text{ReLU}(-2) = 0, \quad q_2 = \text{ReLU}(-2) = 0$$
$$s = (1)(0) + (-2)(0) = 0, \qquad \boxed{o = 0}$$

(c) Numerical Gradient [4 pts]

$$\frac{\partial \mathcal{L}}{\partial o} = o - y = 0 - (-2) = 2$$

$p_2 = -2 < 0 \;\Rightarrow\; \mathbf{1}[p_2 > 0] = 0$

$$\frac{\partial \mathcal{L}}{\partial w_{22}} = 2 \cdot (-2) \cdot 0 \cdot 2 = \boxed{0}$$

(e) SLP Equivalence (linear ϕ_1) [5 pts]

$$\mathbf{v} = \mathbf{W}^\top \mathbf{u} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} (1)(1) + (1)(-2) \\ (-1)(1) + (-1)(-2) \end{bmatrix} = \begin{bmatrix} 1-2 \\ -1+2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

SLP weights: $v_1 = -1$, $v_2 = 1$. Output: $o = -x_1 + x_2$.

Rubric Summary for Q2

Part	Marks	Criteria
(a) Forward pass	3	1 pt per stage: pre-activations p_1, p_2; ReLU outputs q_1, q_2; final output o.
(b) Chain rule	3	All 5 terms in correct chain (1 pt for $\partial \mathcal{L}/\partial o$; 1 pt for $\partial s/\partial q_2$ and $\partial q_2/\partial p_2$; 1 pt for $\partial p_2/\partial w_{22}$). Omitting trivially-1 term ($\partial o/\partial s = 1$) loses 0.5 pt.
(c) Numerical grad	4	1 pt: correct $o - y$; 1 pt: correct ReLU state; 1 pt: correct u_2 and x_2 values; 1 pt: correct final number. For V2/V4 where gradient is 0, deduct 2 pts if student ignores dead neuron.
(d)/(e) SLP	5	2 pts: correct argument that linear MLP collapses to SLP; 2 pts: correct matrix calculation $\mathbf{v} = \mathbf{W}^\top \mathbf{u}$; 1 pt: correct diagram with numerical weights.

3 Question 3 — Single Layer Perceptron Classification [15 pts]

Part (a) — Classification Rule [3 pts]

Classification rule for all variants:

$$\hat{y} = \operatorname{sign}(w_0 + w_1 x_1 + w_2 x_2)$$

where $\operatorname{sign}(z) = +1$ if $z \geq 0$, and -1 if $z < 0$.

The activation function is the **sign function** (Heaviside / threshold activation).

Some sources define $\operatorname{sign}(0) = -1$ (strict threshold). **Both are acceptable** provided the student is consistent. This only affects samples with exactly zero net input.

Variant 1: $D = \{([1,1]^T, +1),\ ([0,0]^T, +1),\ ([-1,-1]^T, -1)\}$; $w_1 = -0.5,\ w_2 = -0.5,\ w_0 = 0$

$\mathbf{x}$	y	$w_0 + w_1x_1 + w_2x_2$	Pred $\hat{y}$	Error?
$[1,1]$	$+1$	$0 - 0.5 - 0.5 = -1$	-1	**Yes**
$[0,0]$	$+1$	$0 + 0 + 0 = 0$	$+1$	No
$[-1,-1]$	-1	$0 + 0.5 + 0.5 = 1$	$+1$	**Yes**

Error on training set: 2 out of 3 samples.

If student uses sign(0) $= -1$: $[0,0]$ predicts -1, error. Total errors $= 3/3$. Accept with consistent sign convention.

Variant 2: $D = \{([1,1]^T, -1),\ ([0,0]^T, -1),\ ([-1,-1]^T, +1)\};\quad w_1 = 0.5,\ w_2 = 0.5,\ w_0 = -0.5$

$\mathbf{x}$	y	$w_0 + w_1x_1 + w_2x_2$	Pred $\hat{y}$	Error?
$[1,1]$	-1	$-0.5 + 0.5 + 0.5 = 0.5$	$+1$	**Yes**
$[0,0]$	-1	$-0.5 + 0 + 0 = -0.5$	-1	No
$[-1,-1]$	$+1$	$-0.5 - 0.5 - 0.5 = -1.5$	-1	**Yes**

Error on training set: 2 out of 3 samples.

Variant 3: $D = \{([1,1]^T, +1),\ ([0,0]^T, +1),\ ([-1,-1]^T, -1)\};\quad w_1 = -0.8,\ w_2 = -0.8,\ w_0 = 0$

$\mathbf{x}$	y	$w_0 + w_1x_1 + w_2x_2$	Pred $\hat{y}$	Error?
$[1,1]$	$+1$	$0 - 0.8 - 0.8 = -1.6$	-1	**Yes**
$[0,0]$	$+1$	$0 + 0 + 0 = 0$	$+1$	No
$[-1,-1]$	-1	$0 + 0.8 + 0.8 = 1.6$	$+1$	**Yes**

Error on training set: 2 out of 3 samples.

Variant 4: $D = \{([1,1]^T, -1),\ ([0,0]^T, +1),\ ([-1,-1]^T, +1)\};\quad w_1 = 0.7,\ w_2 = 0.7,\ w_0 = 0.5$

$\mathbf{x}$	y	$w_0 + w_1x_1 + w_2x_2$	Pred $\hat{y}$	Error?
$[1,1]$	-1	$0.5 + 0.7 + 0.7 = 1.9$	$+1$	**Yes**
$[0,0]$	$+1$	$0.5 + 0 + 0 = 0.5$	$+1$	No
$[-1,-1]$	$+1$	$0.5 - 0.7 - 0.7 = -0.9$	-1	**Yes**

Error on training set: 2 out of 3 samples.

Rubric for (a):

- 1 pt: correct rule with sign/step activation function stated.
- 1 pt: correct output on all three samples (consistent convention).
- 1 pt: correct error count.

Part (b) — Perceptron Training Algorithm [4 pts]

Perceptron Learning Algorithm:

Initialize: weights $\mathbf{w} = [w_0, w_1, w_2]^\top$, learning rate $\eta > 0$.

Repeat until convergence (or for T epochs):

1. For each training example $(\mathbf{x}_i, y_i)$ in the dataset:

   a. Compute: $a_i = w_0 + w_1x_{i1} + w_2x_{i2}$

b. Predict: $\hat{y}_i = \text{sign}(a_i)$
c. **If** $\hat{y}_i \neq y_i$ (misclassified):

$$w_j \leftarrow w_j + \eta\, y_i\, x_{ij} \quad \forall j \in \{1, 2\}$$
$$w_0 \leftarrow w_0 + \eta\, y_i$$

Rubric for (b):

- 1 pt: update rule written correctly ($\mathbf{w} \leftarrow \mathbf{w} + \eta y_i \mathbf{x}_i$).
- 1 pt: condition for update is "misclassified" ($\hat{y}_i \neq y_i$).
- 1 pt: bias update $w_0 \leftarrow w_0 + \eta y_i$ included.
- 1 pt: loop structure (iterate over samples, repeat until convergence).

Part (c) — Two Perceptron Iterations ($\eta = 0.5$) [8 pts]

Update rule: For misclassified $(\mathbf{x}, y)$: $w_j \leftarrow w_j + \eta\, y\, x_j, \quad w_0 \leftarrow w_0 + \eta\, y\, (1)$ (augmented bias input $= 1$).
For correct samples: **no update**.

Variant 1 Perceptron Iterations

Initial: $w_0 = 0,\ w_1 = -0.5,\ w_2 = -0.5$

Iteration 1:

Sample	Computation	Update
$([1, 1], +1)$	$a = 0 + (-0.5)(1) + (-0.5)(1) = -1 < 0 \Rightarrow \hat{y} = -1 \neq +1$. **Misclassified.**	$w_0 \leftarrow 0 + 0.5(+1) = 0.5$ $w_1 \leftarrow -0.5 + 0.5(+1)(1) = 0$ $w_2 \leftarrow -0.5 + 0.5(+1)(1) = 0$
$([0, 0], +1)$	$a = 0.5 + 0 + 0 = 0.5 \geq 0 \Rightarrow \hat{y} = +1 = y$. Correct.	No update.
$([-1, -1], -1)$	$a = 0.5 + (0)(-1) + (0)(-1) = 0.5 \geq 0 \Rightarrow \hat{y} = +1 \neq -1$. **Misclassified.**	$w_0 \leftarrow 0.5 + 0.5(-1) = 0$ $w_1 \leftarrow 0 + 0.5(-1)(-1) = 0.5$ $w_2 \leftarrow 0 + 0.5(-1)(-1) = 0.5$

End of Iteration 1: $w_0 = 0,\ w_1 = 0.5,\ w_2 = 0.5$

Iteration 2:

Sample	Computation	Update
$([1, 1], +1)$	$a = 0 + 0.5 + 0.5 = 1 \geq 0 \Rightarrow +1 = y$. Correct.	No update.
$([0, 0], +1)$	$a = 0 \geq 0 \Rightarrow +1 = y$. Correct.	No update.
$([-1, -1], -1)$	$a = 0 - 0.5 - 0.5 = -1 < 0 \Rightarrow -1 = y$. Correct.	No update.

End of Iteration 2: $w_0 = 0,\ w_1 = 0.5,\ w_2 = 0.5$. **Converged** (all samples correctly classified).

Variant 2 Perceptron Iterations

Initial: $w_0 = -0.5,\ w_1 = 0.5,\ w_2 = 0.5$

Iteration 1:

Sample	Computation	Update
$([1,1],-1)$	$a=-0.5+0.5+0.5=0.5\geq 0\Rightarrow +1\neq -1$. **Misclassified.**	$w_0\leftarrow -0.5+0.5(-1)=-1$ $w_1\leftarrow 0.5+0.5(-1)(1)=0$ $w_2\leftarrow 0.5+0.5(-1)(1)=0$
$([0,0],-1)$	$a=-1+0+0=-1<0\Rightarrow -1=y$. Correct.	No update.
$([-1,-1],+1)$	$a=-1+(0)(-1)+(0)(-1)=-1<0\Rightarrow -1\neq +1$. **Misclassified.**	$w_0\leftarrow -1+0.5(+1)=-0.5$ $w_1\leftarrow 0+0.5(+1)(-1)=-0.5$ $w_2\leftarrow 0+0.5(+1)(-1)=-0.5$

End of Iteration 1: $w_0=-0.5,\ w_1=-0.5,\ w_2=-0.5$

Iteration 2:

Sample	Computation	Update
$([1,1],-1)$	$a=-0.5-0.5-0.5=-1.5<0\Rightarrow -1=y$. Correct.	No update.
$([0,0],-1)$	$a=-0.5<0\Rightarrow -1=y$. Correct.	No update.
$([-1,-1],+1)$	$a=-0.5+0.5+0.5=0.5\geq 0\Rightarrow +1=y$. Correct.	No update.

End of Iteration 2: $w_0=-0.5,\ w_1=-0.5,\ w_2=-0.5$. **Converged.**

Variant 3 Perceptron Iterations

Initial: $w_0=0,\ w_1=-0.8,\ w_2=-0.8$

Iteration 1:

Sample	Computation	Update
$([1,1],+1)$	$a=0-0.8-0.8=-1.6<0\Rightarrow -1\neq +1$. **Misclassified.**	$w_0\leftarrow 0+0.5(+1)=0.5$ $w_1\leftarrow -0.8+0.5(+1)(1)=-0.3$ $w_2\leftarrow -0.8+0.5(+1)(1)=-0.3$
$([0,0],+1)$	$a=0.5\geq 0\Rightarrow +1=y$. Correct.	No update.
$([-1,-1],-1)$	$a=0.5+(-0.3)(-1)+(-0.3)(-1)=0.5+0.3+0.3=1.1\geq 0\Rightarrow +1\neq -1$. **Misclassified.**	$w_0\leftarrow 0.5+0.5(-1)=0$ $w_1\leftarrow -0.3+0.5(-1)(-1)=0.2$ $w_2\leftarrow -0.3+0.5(-1)(-1)=0.2$

End of Iteration 1: $w_0=0,\ w_1=0.2,\ w_2=0.2$

Iteration 2:

Sample	Computation	Update
$([1,1],+1)$	$a = 0 + 0.2 + 0.2 = 0.4 \geq 0 \Rightarrow +1 = y$. Correct.	No update.
$([0,0],+1)$	$a = 0 \geq 0 \Rightarrow +1 = y$. Correct.	No update.
$([-1,-1],-1)$	$a = 0 - 0.2 - 0.2 = -0.4 < 0 \Rightarrow -1 = y$. Correct.	No update.

End of Iteration 2: $w_0 = 0,\ w_1 = 0.2,\ w_2 = 0.2$. **Converged.**

Variant 4 Perceptron Iterations

Initial: $w_0 = 0.5,\ w_1 = 0.7,\ w_2 = 0.7$

Iteration 1:

Sample	Computation	Update
$([1,1],-1)$	$a = 0.5 + 0.7 + 0.7 = 1.9 \geq 0 \Rightarrow +1 \neq -1$. **Misclassified.**	$w_0 \leftarrow 0.5 + 0.5(-1) = 0$ $w_1 \leftarrow 0.7 + 0.5(-1)(1) = 0.2$ $w_2 \leftarrow 0.7 + 0.5(-1)(1) = 0.2$
$([0,0],+1)$	$a = 0 \geq 0 \Rightarrow +1 = y$. Correct.	No update.
$([-1,-1],+1)$	$a = 0 + (0.2)(-1) + (0.2)(-1) = -0.4 < 0 \Rightarrow -1 \neq +1$. **Misclassified.**	$w_0 \leftarrow 0 + 0.5(+1) = 0.5$ $w_1 \leftarrow 0.2 + 0.5(+1)(-1) = -0.3$ $w_2 \leftarrow 0.2 + 0.5(+1)(-1) = -0.3$

End of Iteration 1: $w_0 = 0.5,\ w_1 = -0.3,\ w_2 = -0.3$

Iteration 2:

Sample	Computation	Update
$([1,1],-1)$	$a = 0.5 - 0.3 - 0.3 = -0.1 < 0 \Rightarrow -1 = y$. Correct.	No update.
$([0,0],+1)$	$a = 0.5 \geq 0 \Rightarrow +1 = y$. Correct.	No update.
$([-1,-1],+1)$	$a = 0.5 + 0.3 + 0.3 = 1.1 \geq 0 \Rightarrow +1 = y$. Correct.	No update.

End of Iteration 2: $w_0 = 0.5,\ w_1 = -0.3,\ w_2 = -0.3$. **Converged.**

Rubric for Q3(c) [8 pts]

Marks	Criteria
2	Correctly identifies all misclassified samples in Iteration 1 (consistent with initial weights and chosen sign(0) convention).
2	Correct updates applied only to misclassified samples using $\Delta w_j = \eta\, y\, x_j$.
2	Correct weights after Iteration 1.
1	Correct Iteration 2 classification using updated weights.
1	Correct final weights after Iteration 2 / notes convergence.
Deductions:	
-1	Updating correct samples (i.e. updating all samples regardless of classification).
-1	Wrong sign on update (e.g. $w \leftarrow w - \eta y x$ instead of $+$).
-0.5	Minor arithmetic error (carry through if method is correct).
-1	Wrong learning rate applied.

4 Quick Reference: Final Answers by Variant

	V1	**V2**	**V3**	**V4**
Q1a(i)	F	F	F	F
Q1a(ii)	F	T	F	T
Q1a(iii)	T	T	F	F
Q1c μ	10	20	50	100
Q1c σ **(pop)**	2	4	5	5
Q1c σ **(samp)**	$\sqrt{4.8}$	$\sqrt{19.2}$	$\sqrt{30}$	$\sqrt{30}$
Q1e params	20	18	24	40
Q2(a) output o	6	0	2	0
Q2 $\partial\mathcal{L}/\partial w_{22}$	4	0 (or -12)	2	0
Q2 SLP v_1	3	-2	-6	-1
Q2 SLP v_2	3	2	4	1
Q3(a) errors	2/3	2/3	2/3	2/3
Q3 final w_0	0	-0.5	0	0.5
Q3 final w_1	0.5	-0.5	0.2	-0.3
Q3 final w_2	0.5	-0.5	0.2	-0.3
Q3 converged?	Yes (iter 2)	Yes (iter 2)	Yes (iter 2)	Yes (iter 2)